

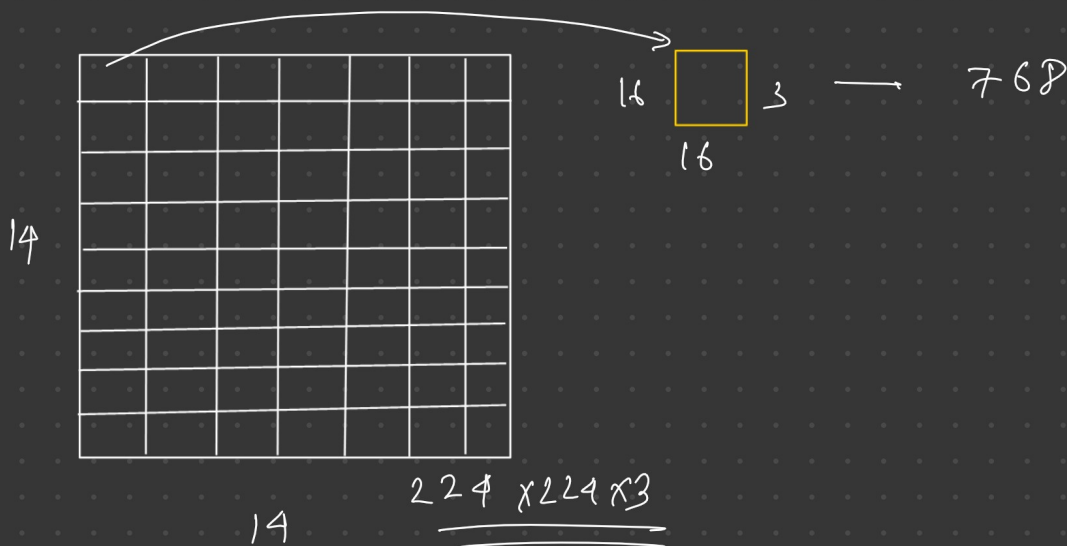
Agenda - 01 - March - 2025

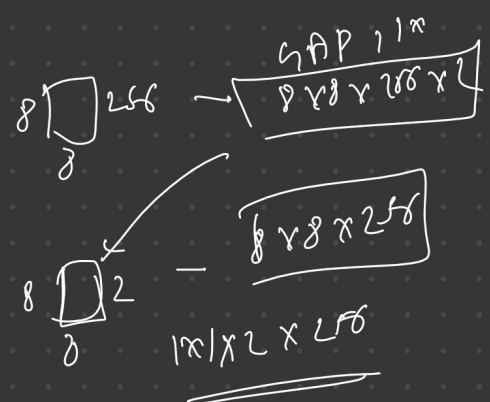
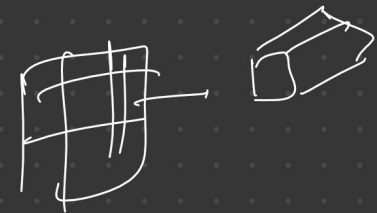
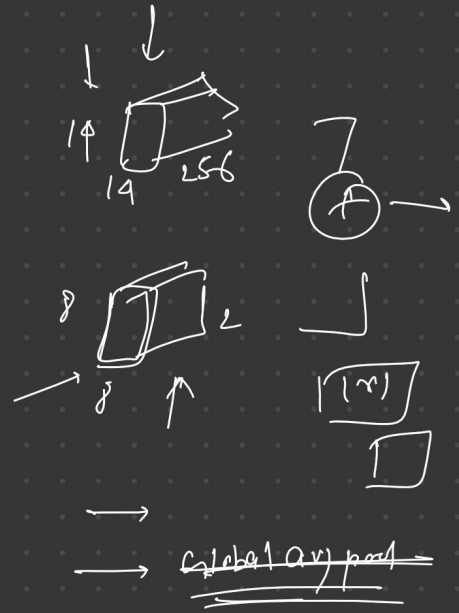
- Vi ↑
- IOU function (bounding box) - IOU, GIoU, DIOU, CIOU
- Detector → licence

ViT → An image is worth 16×16 words

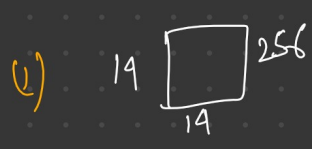
$$\begin{array}{l} 224 / 16 \rightarrow \underline{14} \\ 224 / 16 \rightarrow 14 \end{array} \rightarrow 14 \times 14, \text{ Each patch having } \underline{16 \times 16 \text{ pixels}}$$

$$14 \times 14 \rightarrow 196 \text{ patches} \rightarrow \boxed{16 \times 16 \times 3}$$





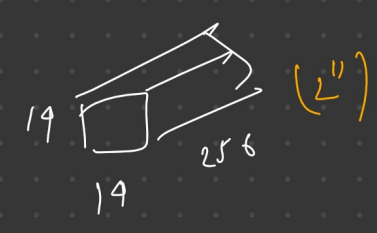
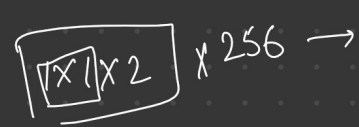
→ (1) Upsample (2)



2 → 256



→ use 1x1 conv



(1) + (2'') → $[14 \times 14 \times 256]$

ML →

linear regression → find the best
least square pattern → best fit
line

→ $y = mx + c$

→ $= m_1 x_1 + m_2 x_2 + c$

→ least error

$m = 2$

↓

1.99
1.98

↓	1	2	←
↓	2	4	←
↓	3	6	←
↓	4	8	←
↓	5	10	←

ML, DL, CV, RF, RL → var

→ Image data → 1m 1000

640

↓ n.p. a

77	20	69
45	99	30
	20	

→ 97 → avg = 640 x 12

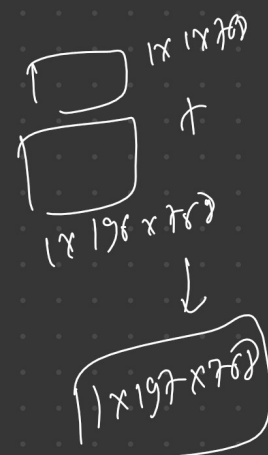
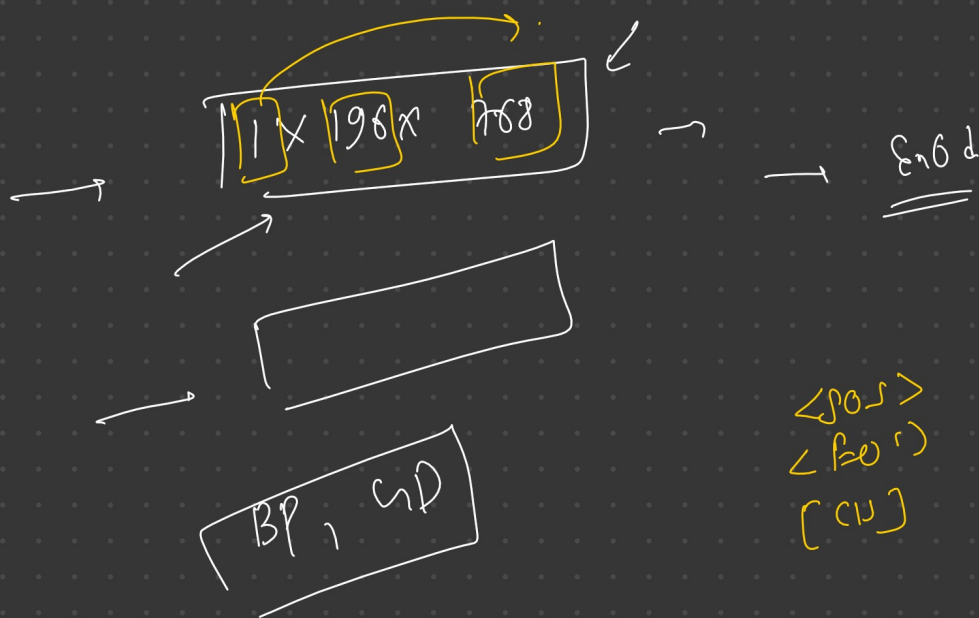
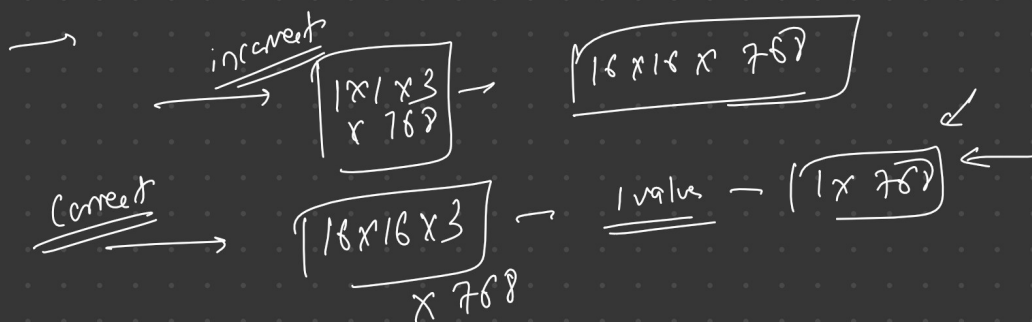
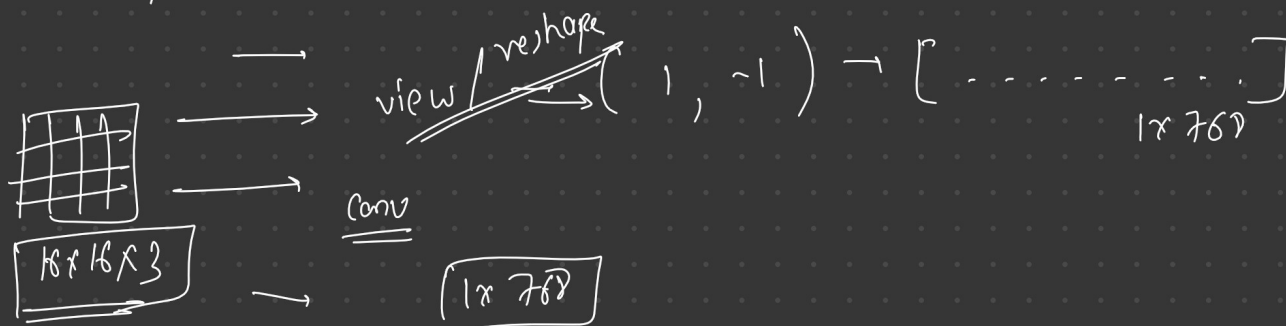
↓

ar

encoding → word2vec, fasttext, GloVe

sentence → encoded

→ $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1024 \end{bmatrix}$
↑ ↑ ↑ ↑ ↑ ↑



<pos>
<word>
[cu]

32×32

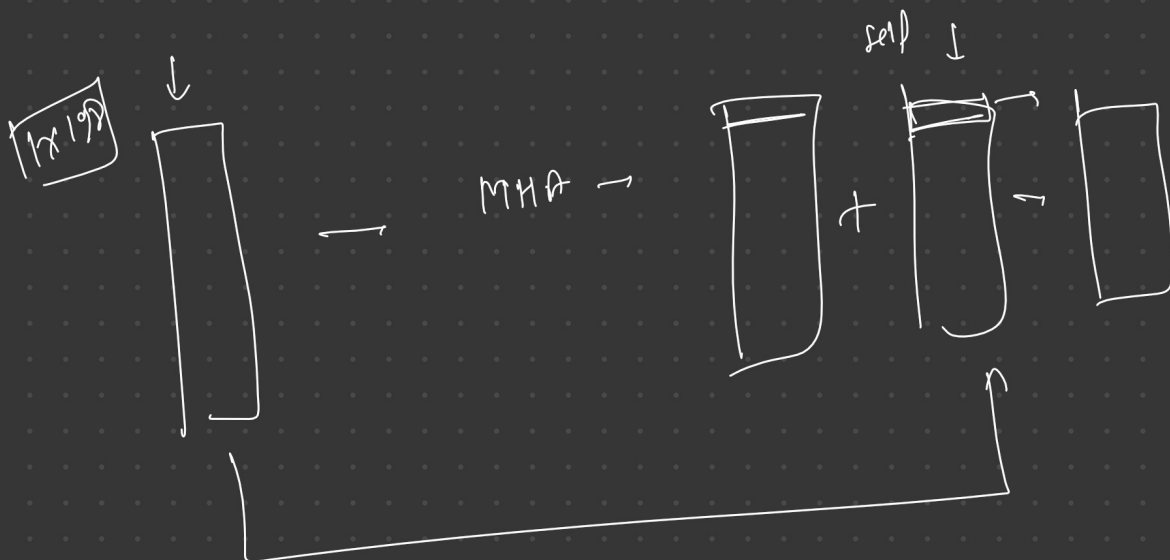
- $\square$

$4 \times 9 \times 3$ piles

8×8

128, 3, 32, 32

128, 3, 8, 4, 8, 4



PCR
pos cos $\rightarrow$
pos sin $\rightarrow$

1×768

16×768

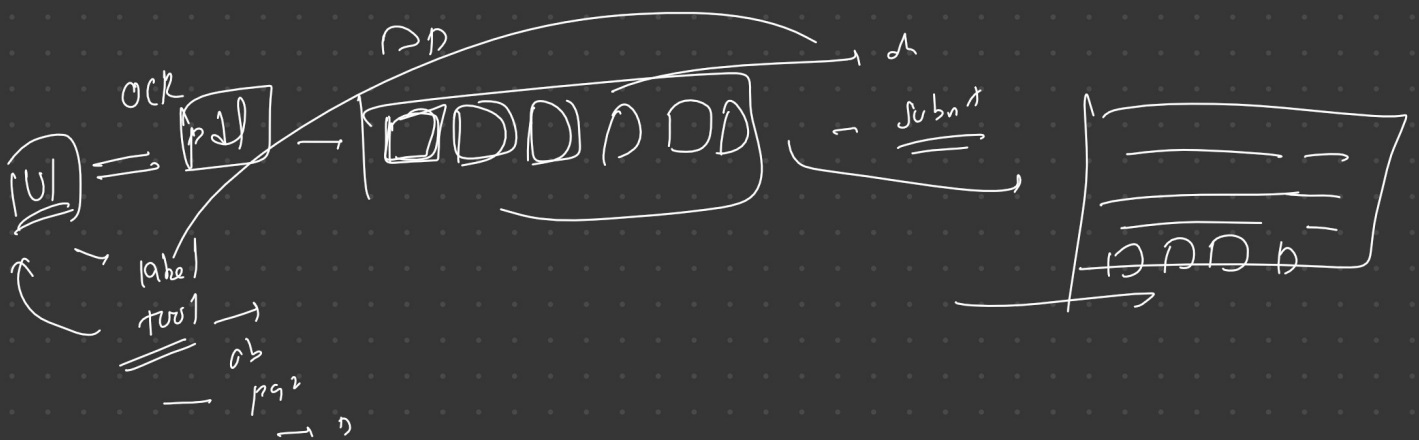
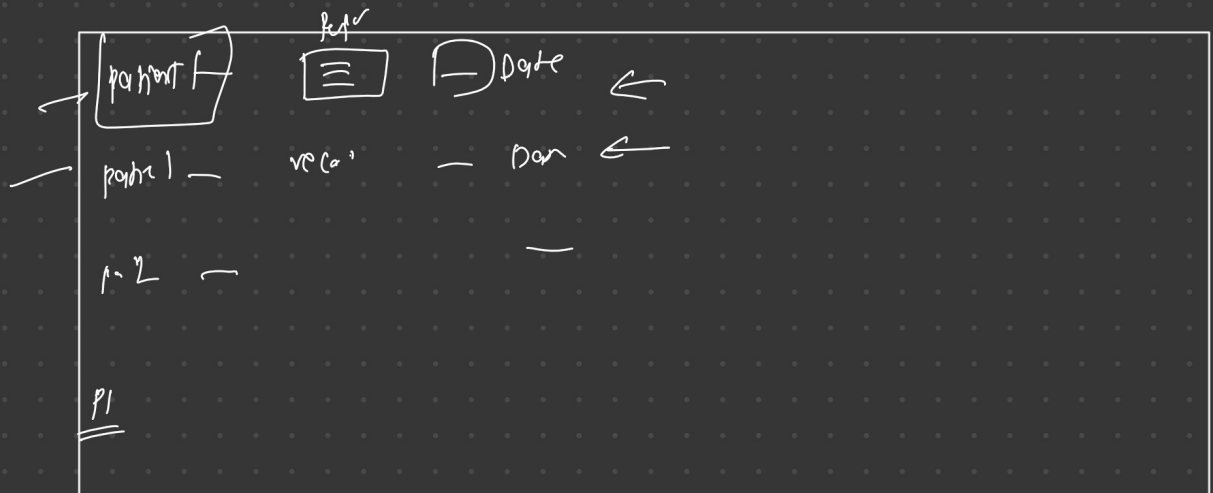
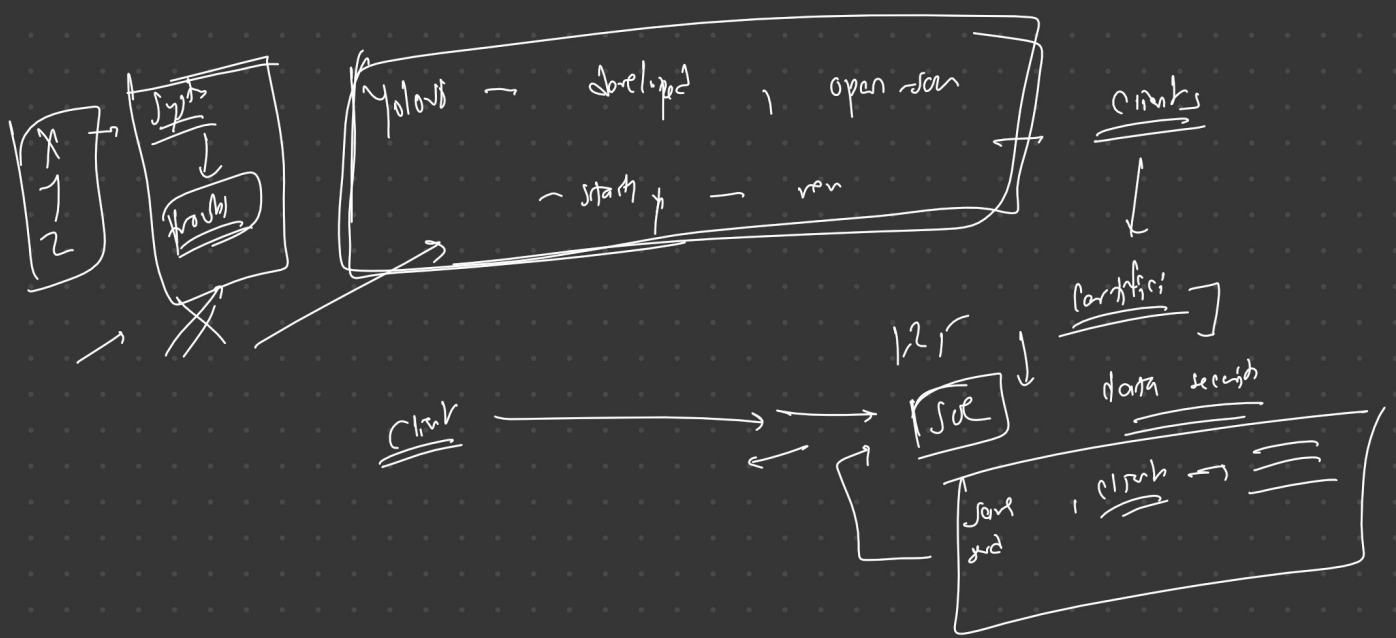
16 $\rightarrow$

nm, rm

fr.

opm

opm, $\square$



σ $\downarrow$ AP^1